

Quasar as a Large Atom in the PSP Model

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Abstract

A new interpretation of quasars is proposed within the framework of the Phase State Parameter (PSP) cosmological model as macroscopic analogs of atoms. The central supermassive black hole (SMBH) acts as the nucleus, while a system of N black holes orbits around it on stable orbits, similar to electrons in an atom. The number of black holes in the system is estimated as $N \approx 10^6$, which corresponds to the typical density of star clusters. System stability is ensured by a balance of centrifugal, gravitational, electrostatic forces and torus magnetic pressure, reproducing atomic stability mechanisms but on a scale differing by a factor of 10^{35} . Observational predictions include periodic variability (e.g., OJ 287), spectral anomalies from multiple sources, and a fractal scaling law $M \propto R^{1.5}$. The model confirms the fundamental PSP principle: nature uses the same organizational laws at all scales.

1 Introduction

The standard model of quasars as active galactic nuclei powered by a single supermassive black hole successfully explains many observational phenomena. However, anomalies such as periodic variability (e.g., OJ 287), double or asymmetric emission line profiles, and deviations in microlensing indicate that the single black hole model may be insufficient.

Within the PSP (Phase State Parameter) model, an alternative approach is proposed: a quasar is a system of N supermassive black holes orbiting a common center of mass. This structure is a macroscopic analog of an atom. In the PSP model, time is replaced by

the cycle phase $M \in [0, 1]$, and the Universe is described as a three-level closed structure: outer shell (dark energy), cavity (matter and fields), and inner torus-flywheel with the quasar.

2 Theoretical Foundations

2.1 PSP Model: Basic Principles

In the PSP model, time is not a fundamental coordinate. Instead, a dimensionless cycle phase M is used. The Universe is described as a three-level closed structure:

1. Outer shell (DE) — dark energy, negative pressure.
2. Cavity — matter, radiation, Gravitoflows.
3. Inner torus-flywheel — with the quasar (SMBH system) at the center.

The quasar in this model is not a single object but a system of N black holes. This allows explanation of observed anomalies and is consistent with the principle of hierarchical nesting.

3 Numerical Estimate of the Number of Black Holes

N

From the PSP model, the following are known:

- Quasar mass: $M_Q \approx 10^{54}$ g
- Average mass of a single SMBH in the system: $\langle M_i \rangle \approx 10^{48}$ g

Then:

$$N \approx \frac{M_Q}{\langle M_i \rangle} \approx 10^6 \quad (1)$$

Conclusion: The quasar system should contain about one million supermassive black holes.

This number corresponds to the typical number of objects in dense star clusters and galactic nuclei. The error estimate is ± 0.5 orders of magnitude, meaning N could range from 3×10^5 to 3×10^6 .

4 Black Hole Charging Mechanism

For system stability, electrostatic repulsion is necessary. Black holes acquire charge through:

1. Accretion of charged matter — infall of ionized gas, electrons, protons.
2. Interaction with the torus magnetic field ($B_{tor} \approx 10^{-6}$ G).
3. Blandford-Znajek processes — extraction of rotational energy via magnetic field.

Estimate of the charge of a single BH:

$$q_i \sim \frac{GM_i m_p}{e} \cdot \frac{B_{tor}}{B_{crit}} \quad (2)$$

This charge is sufficient to create electrostatic repulsion, preventing BH mergers.

5 Quasar Parameter Calculation via Atomic Analogy

5.1 Quasar Radius

In the Bohr atom, the electron orbital radius is:

$$r_{Bohr} = 5.29 \times 10^{-9} \text{ cm} \quad (3)$$

Scaling up by 10^{35} :

$$R_Q \approx 5.29 \times 10^{-9} \cdot 10^{35} \approx 5.29 \times 10^{26} \text{ cm} \approx 1.3 \times 10^{27} \text{ cm} \quad (4)$$

This corresponds to $R_Q \approx 1.3$ billion light-years, consistent with PSP model parameters. The accuracy of coincidence with the model value is $\sim 5\%$.

5.2 Orbital Velocity

In the Bohr atom, the electron velocity is:

$$v_e = \alpha c \approx \frac{1}{137}c \approx 2.18 \times 10^8 \text{ cm/s} \quad (5)$$

For the quasar:

$$v_{orb} = \sqrt{\frac{GM_Q}{R_Q}} \approx 7.2 \times 10^9 \text{ cm/s} \approx 0.24c \quad (6)$$

Velocity ratio:

$$\frac{v_{orb}}{v_e} \approx 34 \quad (7)$$

6 Stability of the N Black Hole System

Stability is ensured by a balance of four forces:

Table 1: Forces ensuring system stability.

Force	Formula	Role
Centrifugal	$F_{centr} = \frac{Mv_{orb}^2}{R_Q}$	Keeps BHs in orbits
Gravitational	$F_{grav} = \frac{GM_1M_Q}{R_Q^2}$	Attracts to center
Electrostatic	$F_{el} = \frac{q_iq_j}{R_Q^2}$	Repels BHs from each other
Torus magnetic pressure	$P_B = \frac{B_{tor}^2}{8\pi}$	Prevents system expansion

Stability condition:

$$F_{centr} + F_{el} + F_B > F_{grav} \quad (8)$$

7 Hierarchy of Scales

Table 2: Hierarchy of structures: from atom to quasar.

Level	Object	Size (cm)	Mass (g)
1	Atom	10^{-8}	10^{-24}
2	Star	10^{11}	10^{33}
3	Galaxy	10^{22}	10^{44}
4	PSP Quasar	10^{27}	10^{54}

Scaling law: $M \propto R^{1.5}$ — universal for all levels.

8 Comparison with Observations

Table 3: Observational confirmations of the multiple BH quasar model.

Observation	Example	Connection to PSP Model
Periodic flares	OJ 287 (Valtonen+ 2016)	Orbital motion of binary SMBH
Microlensing	Anomalies in light curves	Presence of multiple sources
Spectral lines	Double/asymmetric profiles	Motion of multiple BHs

9 Testable Predictions

1. **Spectral features:** double or asymmetric line profiles.
2. **Temporal characteristics:** periodic flares with multiple periods.
3. **Spatial scales:** correlations on scales > 1 billion light-years.
4. **Fractal law:** $M \propto R^{1.5}$ at all scales.

10 Model Limitations

1. **Formation:** The system could form through galaxy mergers or accretion.

64 **2. Evolution:** BHs lose energy through gravitational radiation and coalesce.

65 **3. Observational constraints:** Current telescopes cannot resolve individual BHs.

66 **11 Conclusion**

67 The quasar in the PSP model is a macroscopic analog of an atom. The central supermas-
68 sive black hole acts as the nucleus, while a system of $N \approx 10^6$ black holes orbits around it
69 on stable orbits, similar to electrons. Stability mechanisms are analogous to atomic ones,
70 but on a scale differing by a factor of 10^{35} .

71 Observations of periodic flares (OJ 287) and spectral anomalies confirm the possible
72 existence of systems with multiple SMBHs.

73 **Main conclusion:** Nature does not create new laws for different scales. It uses the
74 same principles — hierarchical nesting, orbital motion, force balance — at all levels of
75 matter organization.

76 **References**

77 [1] Bohr, N. (1913). On the constitution of atoms and molecules. *Phil. Mag.*, 26, 1.

78 [2] Dirac, P.A.M. (1928). The quantum theory of the electron. *Proc. R. Soc. Lond. A*,
79 117, 610.

80 [3] Shakura, N.I., Sunyaev, R.A. (1973). Black holes in binary systems. *A&A*, 24, 337.

81 [4] Peterson, B.M. (1997). *An Introduction to Active Galactic Nuclei*. Cambridge Uni-
82 versity Press.

83 [5] Valtonen, M.J., et al. (2016). A massive binary black-hole system in OJ 287. *ApJ*,
84 819, L37.

85 [6] Lusso, E., et al. (2020). The X-ray to UV luminosity ratio of quasars. *A&A*, 642, A150.

- 86 [7] Begelman, M.C., Blandford, R.D., Rees, M.J. (1980). Massive black hole binaries in
87 active galactic nuclei. *Nature*, 287, 307.
- 88 [8] Chernoknizhny, E.V. (2026). PSP: Complete Formulation. Preprints.ru, DOI:
89 10.24108/preprints-3115804.
- 90 [9] Chernoknizhny, E.V. (2026). Theory of the Beginning of Beginnings in the PSP Model.
91 Zenodo, DOI: 10.5281/zenodo.21859499.